



ALLIANCE
UNIVERSITY

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GRADE A+
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COURSE: BCA(HONS)

SUBJECT: DATABASE MANAGEMENT SYSTEM

TOPIC: ONE REAL WORLD APPLICATION IDENTIFIES
THE DATABASE

NAME OF THE APPLICATION: HOSPITAL MANAGEMENT SYSTEM

1. PURPOSE:

- A Hospital Management System is a database-driven application used to manage patients, doctors, appointments, medical records, treatments, prescriptions, rooms, and billing.
- The main goal is to keep hospital information organized and connected so that staff can quickly access accurate patient medical information.

2. REQUIRED DATABASE TABLES:

- For a practical Hospital Management System, the following tables are sufficient:
 1. Patients
 2. Doctors
 3. Departments
 4. Appointments
 5. Medical Records
 6. Prescriptions
 7. Prescription Medicines
 8. Medicines
 9. Admissions
 10. Rooms
 11. Bills
 12. Bill Payments

3.COLUMNS/ATTRIBUTES FOR EACH TABLE:

- TABLE 1: PATIENTS

COLUMN	DESCRIPTION	KEY
Patient ID	Unique patient identifier	Primary Key
First Name	Patient's first name	
Last Name	Patient's last name	
Date of Birth	Patient's date of birth	
Gender	Patient's gender	
Phone	Contact number	
Address	Residential address	
Emergency Contact	Emergency contact number	

Purpose: Stores the basic information of every patient registered with the hospital.

- **TABLE 2: DOCTORS**

COLUMN	DESCRIPTION	KEY
Doctor ID	Unique doctor identifier	Primary Key
FirstName	Doctor's first name	
Last Name	Doctor's last name	
Specialization	Medical specialization	
Phone	Contact number	
Email	Email address	
Department ID	Department where doctor works	Foreign Key

- **Table 3: Departments**

Column	Description	Key
Department ID	Unique department identifier	Primary Key
Department Name	Name of department	
Location	Department location	

Purpose: Represents hospital departments such as Cardiology, Neurology, Emergency, and Paediatrics.

- **Table 4: Appointments**

Column	Description	Key
Appointment ID	Unique appointment identifier	Primary Key
Patient ID	Patient attending appointment	Foreign Key
Doctor ID	Doctor conducting appointment	Foreign Key
Appointment Date	Date of appointment	
Appointment Time	Time of appointment	
Status	Scheduled, Completed, Cancelled, etc.	
Reason	Reason for visit	

Purpose: Manages appointments between patients and doctors.

- **Table 5: Medical Records**

Column	Description	Key
Record ID	Unique medical record identifier	Primary Key
Patient ID	Patient associated with record	Foreign Key
Doctor ID	Doctor who created the record	Foreign Key
Record Date	Date of examination	
Diagnosis	Doctor's diagnosis	
Symptoms	Patient's symptoms	
Treatment	Recommended treatment	
Notes	Additional medical notes	

Purpose: Stores the patient's medical history, diagnoses, symptoms, and treatments.

- **Table 6: Prescriptions**

Column	Description	Key
Prescription ID	Unique prescription identifier	Primary Key
Patient ID	Patient receiving prescription	Foreign Key
Doctor ID	Doctor issuing prescription	Foreign Key
Prescription Date	Date prescription was issued	
Instructions	General prescription instructions	

Purpose: Stores prescriptions issued by doctors to patients.

- **Table 7: Prescription Medicines**

Column	Description	Key
Prescription ID	Associated prescription	Primary Key / Foreign Key
Medicine ID	Prescribed medicine	Primary Key / Foreign Key
Dosage	Amount per dose	
Frequency	How often medicine is taken	
Duration	Treatment duration	

Purpose: Connects prescriptions with medicines. This table is necessary because one prescription can contain multiple medicines, and the same medicine can appear in many prescriptions.

- **Table 8: Medicines**

Column	Description	Key
Medicine ID	Unique medicine identifier	Primary Key
Medicine Name	Name of medicine	
Manufacturer	Manufacturer name	
Unit Price	Price per unit	
Stock Quantity	Quantity currently available	

Purpose: Maintains information about medicines available in the hospital pharmacy.

- **Table 9: Admissions**

Column	Description	Key
Admission ID	Unique admission identifier	Primary Key
Patient ID	Admitted patient	Foreign Key
Room ID	Assigned room	Foreign Key
Admission Date	Date of admission	
Discharge Date	Date of discharge	
Reason	Reason for admission	
Status	Admitted/Discharged	

Purpose: Tracks patients who require hospitalization and record their room assignments and admission/discharge information.

- **Table 10: Rooms**

Column	Description	Key
Room ID	Unique room identifier	Primary Key
Room Number	Hospital room number	
Room Type	General, Private, ICU, etc.	
Daily Rate	Cost per day	
Status	Available/Occupied/Maintenance	

Purpose: Keeps track of hospital rooms and their availability.

- **Table 11: Bills**

Column	Description	Key
Bill ID	Unique bill identifier	Primary Key
Patient ID	Patient being billed	Foreign Key
Admission ID	Related admission, if applicable	Foreign Key
Bill Date	Date bill was generated	
Total Amount	Total amount due	
Status	Paid/Partially Paid/Unpaid	

Purpose: Records the amount a patient is required to pay for hospital services.

- **Table 12: Bill Payments**

Column	Description	Key
Payment ID	Unique payment identifier	Primary Key
Bill ID	Bill being paid	Foreign Key
Payment Date	Date of payment	
Amount	Amount paid	
Payment Method	Cash, Card, UPI, etc.	
Transaction Reference	Payment reference number	

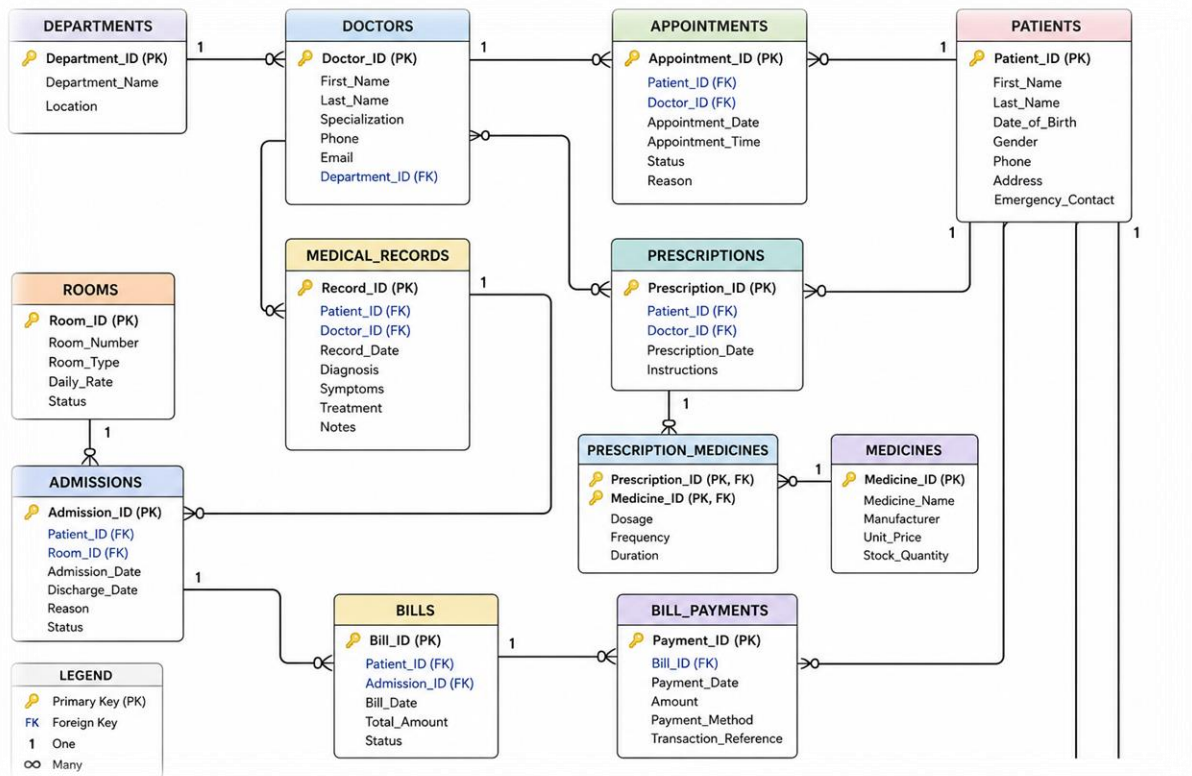
Purpose: Records individual payments made against hospital bills. Multiple payments can be made for one bill.

4.Relationships Between Tables

The main relationships are:

Relationship	Type	Explanation
Departments → Doctors	1: Many	One department can have many doctors.
Patients → Appointments	1: Many	A patient can have many appointments.
Doctors → Appointments	1: Many	A doctor can have many appointments.
Patients → Medical Records	1: Many	A patient can have many medical records.
Doctors → Medical Records	1: Many	A doctor can create many medical records.
Patients → Prescriptions	1: Many	A patient can receive many prescriptions.
Doctors → Prescriptions	1: Many	A doctor can issue many prescriptions.
Prescriptions → Prescription Medicines	1: Many	A prescription can contain several medicines.
Medicines → Prescription Medicines	1: Many	A medicine can appear in many prescriptions.
Patients → Admissions	1: Many	A patient may have multiple hospital admissions over time.
Rooms → Admissions	1: Many	A room can be used for many admissions at different times.
Patients → Bills	1: Many	A patient can have multiple bills.
Admissions → Bills	1: Many	An admission may generate one or more bills.
Bills → Bill Payments	1: Many	A bill can be paid through multiple payments.

5.Simplified Relationship Diagram



6.Overall Explanation

The database represents the hospital as a set of related entities rather than storing everything in one large table.

For example, when a patient books an appointment, the Appointments table connects that patient to a particular doctor. If the doctor diagnoses the patient, the information is stored in Medical Records. If medication is prescribed, the Prescriptions, Prescription Medicines, and Medicines tables work together to record exactly which medicines were prescribed.

If the patient needs to stay in the hospital, an Admission connects the patient to an available Room. Finally, Bills and Bill Payments handle the financial side of the patient's treatment.

This structure reduces unnecessary duplication and makes it possible to answer practical questions such as:

- Which doctor is treating a particular patient?
- Which doctor is treating a particular patient?
- What appointments does a doctor have today?
- What is a patient's medical history?
- Which medicines were prescribed?
- Which rooms are currently occupied?
- How much does a patient owe?
- How much has already been paid?

Thus, the Hospital Management System database provides a realistic example of how a real-world organization can be represented using interconnected database tables